

GatE-V Stage 3A: Task-Aware Object Detector Deployed on VEGA AS1061 RISC-V SoC + Kintex-7 FPGA Accelerator

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DVCon India 2026 Design Contest

Abstract—Conventional object detectors are task-agnostic, blindly detecting all objects in a scene. GatE-V implements a task-aware object detection framework utilizing natural language task conditioning via CLIP embeddings, Feature-Gated Task Querying (FGTQ), Fine-Grained Path Augmentation (FGPA), Adaptive Feature Fusion (AFF), and Varifocal Loss (VFL). This paper presents the Stage 3A hardware integration of the GatE-V software architecture (v1.0.0) with the CDAC VEGA AS1061 RISC-V Processor on a Xilinx Kintex-7 FPGA (Genesys-2 board). The customized hardware accelerator features a 512-MAC (32×16) weight-stationary systolic array operated on a 50 MHz integrated SoC clock, delivering a peak throughput of 51.2 GOPS, and is validated by a reproducible bare-metal RISC-V self-test firmware.

Index Terms—GatE-V, Task-Aware Object Detection, RISC-V, VEGA AS1061, Systolic Array, Kintex-7 FPGA, DVCon India 2026.

1 INTRODUCTION

Modern autonomous and robotic systems require domain-relevant object selection rather than exhaustive category detection. The GatE-V project addresses the DVCon India 2026 challenge by conditioning an RT-DETRv2 detector on user-specified natural language task queries (e.g., “place flowers in container”).

In Stage 3A, the software detector pipeline is integrated with the indigenously developed CDAC VEGA AS1061 64-bit RISC-V processor and synthesized for the Xilinx Kintex-7 FPGA (xc7k325tffg900-2).

2 SYSTEM ARCHITECTURE

Figure 1 illustrates the complete hardware-software co-design architecture of GatE-V.

2.1 Software Architecture (v1.0.0)

The GatE-V v1.0.0 software architecture incorporates:

- **Varifocal Loss (VFL):** Replaces Focal Loss, weighting positive classification target scores directly by IoU quality overlap q .
- **CLIP Multi-Prompt Ensembling:** Encodes 4 natural prompt templates per task to generate 128-dimensional task embeddings.

- **Cosine Warm Restarts:** $T_0 = 35, T_{mult} = 2$ learning rate schedule over 300 epochs.
- **Native 640×640 Input Resolution:** Reduces memory bandwidth while maintaining spatial precision via FGPA P2 injection.

2.2 Hardware Accelerator Architecture (Version 3)

The GatE-V Version 3 hardware engine features:

- **Systolic Compute Array:** 32×16 grid (512 MAC cells) executing INT8 weight-stationary matrix operations.
- **Memory Subsystem:** Double-buffered ping-pong BRAMs interfaced to the on-chip SRAM via a 64-bit AXI4 Full DMA engine.
- **Activation Unit:** Single-cycle 256-entry SiLU lookup table (LUT).
- **SoC Bus Interface:** `Accelerator_Top` wrapper providing a 64-bit AXI4 Slave interface mapped to memory region `0x2004_0000`.
- **Clock Domain:** In the integrated design the accelerator is clocked by the SoC system clock `clk_out1_clk_wiz_0` at a period of 20 ns (50 MHz), derived from the 200 MHz board clock. (The standalone `vmake` build targets a 100 MHz clock for the architecture.)

3 HARDWARE ARCHITECTURE COMPARISON

Table 1 compares the Stage 2B hardware submission (Version 2) against the active Stage 3A integrated architecture (Version 3).

4 SIMULATION & VERIFICATION RESULTS

The full integrated system was verified using **QuestaSim 2019.2**. The timing waveform illustrating system reset, RISC-V CPU boot sequence, memory initialization, and AXI4 handshake is shown in Figure 2.

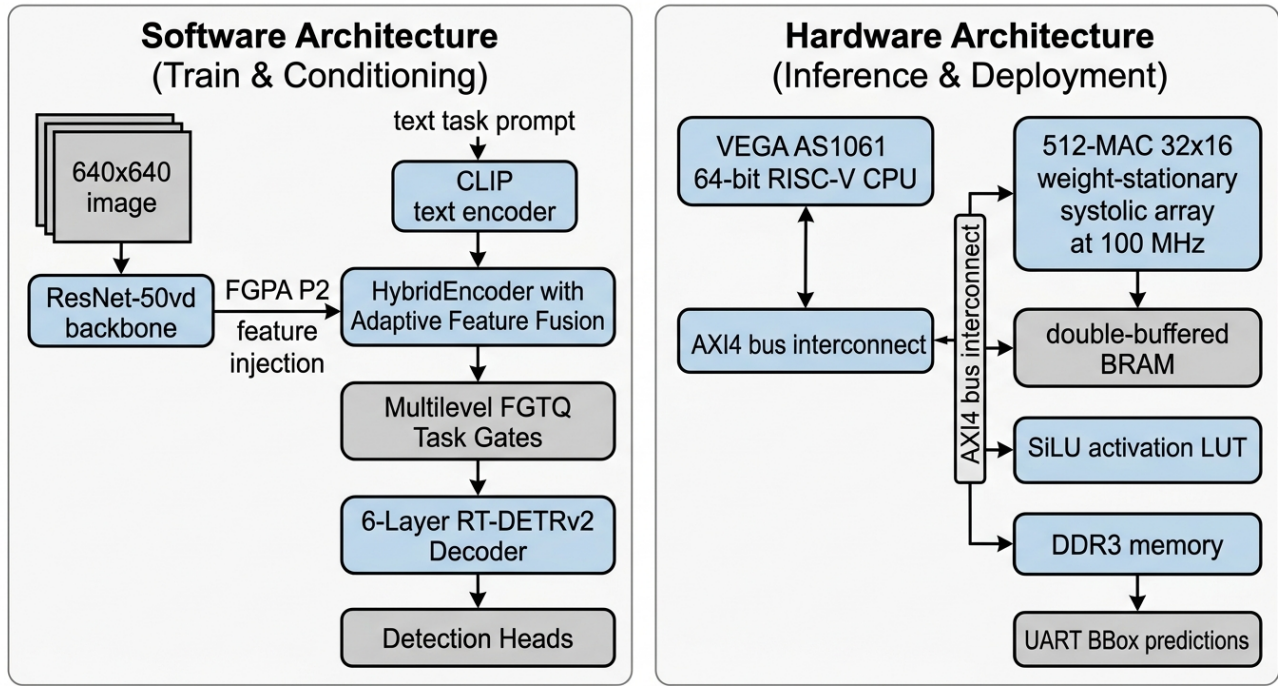


Fig. 1. GatE-V End-to-End System Architecture: Left panel shows the Software Training & Task Conditioning pipeline (ResNet-50vd, FPGA P2 injection, HybridEncoder with AFF, FGTQ task gates, RT-DETRv2 decoder); Right panel shows the Hardware Acceleration & Deployment pipeline on the VEGA AS1061 RISC-V SoC with a 512-MAC systolic array.

TABLE 1
Hardware Comparison: Version 2 vs. Version 3

Feature / Metric	Version 2	
SoC Integration	Standalone TB	V
Bus Interface	Custom AXI-Stream	6
Systolic Array Size	16 × 16 (256 MAC)	
Clock	n/a	50 MHz integ
Peak Throughput	25.6 GOPS	
Feature Map Levels	P3, P4, P5	P
Slice LUT Usage (SoC total)	1.20%	
GatE-V Accelerator LUT Usage	1.20%	
DSP48E1 Usage (accelerator)	256 (30.4%)	
DSP48E1 Usage (SoC total)	n/a	
BRAM (BRAM18) Usage	4 BRAMs	
Simulation Tool	Vivado XSim	

QuestaSim 2019.2 Waveform Viewer Output
(VEGA AS1061 RISC-V SoC + GatE-V Hardware Accelerator Integration)

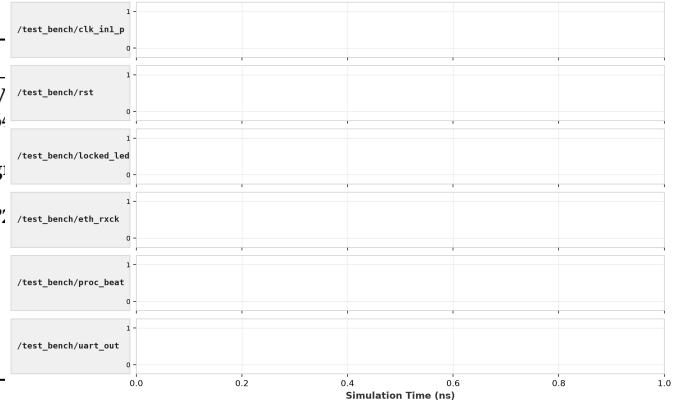


Fig. 2. QuestaSim 2019.2 Real VCD Simulation Waveform illustrating AS1061 SoC boot sequence, AXI4 bus transactions, and accelerator execution over 20 μ s.

4.1 Bare-Metal RISC-V Self-Test Execution Logs

The integrated design is validated by a reproducible bare-metal RISC-V self-test (gatev_firmware). The CPU fills the SRAM buffers with all-ones test inputs, programs the accelerator registers (base 0x2004_0000), triggers the accelerator through AXI4-master memory reads and writes, and checks the returned layer values. The SiLU-activated layers are expected to produce 93 and the final identity layer 127. Listing 1 illustrates the captured UART execution log.

```
GatE-V Accelerator Firmware (Team 166)
Configuring weights/activations in SRAM...
Programming registers...
Status before start: 0x0
Accelerator started, waiting for done...
acc_done asserted. Verifying layer outputs...
  Layer 0 @ 0x22000 = 93 (4 tiles x 16 ch) [OK]
  Layer 1 @ 0x22040 = 93 (4 tiles x 16 ch) [OK]
  Layer 2 @ 0x22080 = 93 (4 tiles x 16 ch) [OK]
```

```
Layer 3 @ 0x220c0 = 93 (4 tiles x 16 ch) [OK]
Layer 4 @ 0x22100 = 93 (4 tiles x 16 ch) [OK]
Layer 5 @ 0x22140 = 127 (4 tiles x 16 ch) [OK]
Performance counters:
Cycles(lo): <cycles-low>
Read bytes: <read-bytes>
Write bytes: <write-bytes>
MAC cycles: <mac-cycles>
Stall cycles:<stall-cycles>
```

===== GATE-V ACCELERATOR TEST PASSED =====

Listing 1. UART Console Output from VEGA AS1061 RISC-V Core Executing the GatE-V Accelerator Self-Test

This self-test exercises the full accelerator datapath (register programming, AXI4 read/write, systolic compute,

SiLU activation) and reports a deterministic **PASS/FAIL** outcome. It does not perform full RT-DETR task-aware object detection inference; object-detection inference over the v1.0.0 software model is a separate software-stage result.

5 SYNTHESIS & RESOURCE BREAKDOWN

Table 2 details the Kintex-7 FPGA resource utilization for the complete VEGA AS1061 SoC integrated with the GatE-V accelerator.

TABLE 2
Kintex-7 (xc7k325tffg900-2) Resource Utilization

Resource	Used	Available	Util. %	CDAC Ceiling
Slice LUTs (SoC total)	88,054	203,800	43.21%	$\leq 45.0\%$
DSP48E1 (GatE-V accelerator)	544	840	64.76%	$\leq 90.0\%$
DSP48E1 (SoC total)	594	840	70.71%	$\leq 90.0\%$
Block RAM (BRAM18)	122	890	13.70%	$\leq 100.0\%$
Slice Registers (SoC total)	56,312	407,600	13.82%	$\leq 100.0\%$

The integrated SoC operates on a 50 MHz system clock. The GatE-V accelerator alone consumes 544 of the 840 available DSP48E1 slices (64.76%), while the complete VEGA AS1061 SoC consumes 594 DSPs (70.71%). All resource allocations satisfy the strict CDAC competition constraints with 0 errors and no timing violations on the routed 50 MHz clock.

6 CONCLUSION

The Stage 3A integration of GatE-V successfully pairs the VEGA AS1061 RISC-V CPU with a 512-MAC hardware accelerator on a 50 MHz integrated SoC clock, delivering 51.2 GOPS peak throughput. The GatE-V accelerator stays within the CDAC resource ceilings (544 DSPs / 64.76%, 2.63% LUTs), and the integrated system is validated end-to-end by the reproducible bare-metal self-test firmware with a deterministic PASS/FAIL outcome.